Case Study:

Design minimum 10 project and product items

Project:	**Product:**
Library Management System	Instagram
Train Reservation System	Whatsapp
College Management System	Netflix
Hotel Management System	Uber
Attendance Management System	Spotify
Airline Booking System	Google Maps
Hostel Management System	BookMyShow
Finance Management System	Meesho
Student Management System	PhonePe
Employee Management System	Gmail

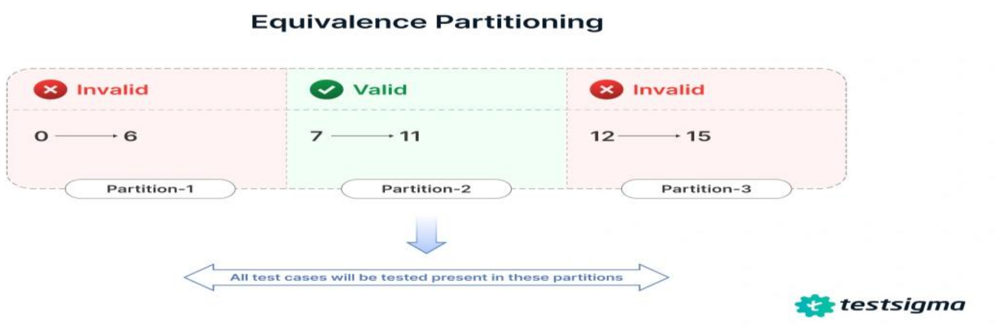
Case Study 3:

Blackbox Test-Case Designed Techniques

1.Equivalence class partitioning(ECP)

Design Equivalence Class Partioning for an application number field that accepts 1-500 digits

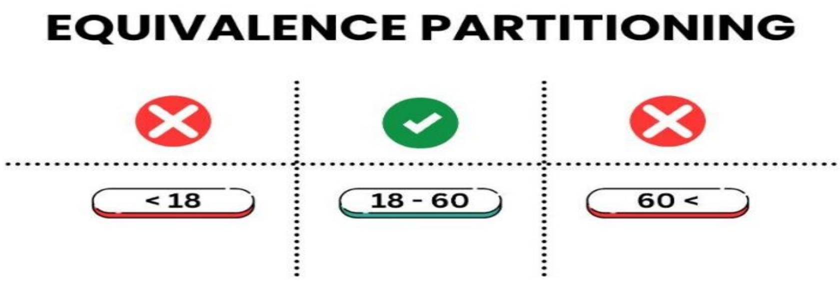
ECP Testing is done by valid and invalid conditions.



S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	Partition 1	0-6->6 (Invalid)	6
2	Partition 2	7-9->8 (Valid)	8
3	Partition 3	12-14->13 (Invalid)	14

Find out Equivalence class partitioning for insurance application which accepts name field, alphabets only.

S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	**Partition 1**	Numbers → 0–9 → **Invalid**	**5**
2	**Partition 2**	Alphabets → A–Z, a–z → **Valid**	**Kirthi**
3	**Partition 3**	Special characters → @, #, $, % → **Invalid**	**@**
4	**Partition 4**	Alphanumeric → A–Z/a–z + 0–9 → **Invalid**	**Kirthi123**
5	**Partition 5**	Space → Space between characters → **Invalid**	**Kirthi Konduru**
6	**Partition 6**	Blank/Empty → No value → **Invalid**	**Blank**



S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	**Partition 1**	Age → **Below 18** → **Invalid**	**17**
2	**Partition 2**	Age → **18–60** → **Valid**	**30**
3	**Partition 3**	Age → **Above 60** → **Invalid**	**61**

Percentage Accepts Percentage value between 50 to 90

Equivalence Partitioning		
Invalid	Valid	Invalid
<=50	50-90	>=90

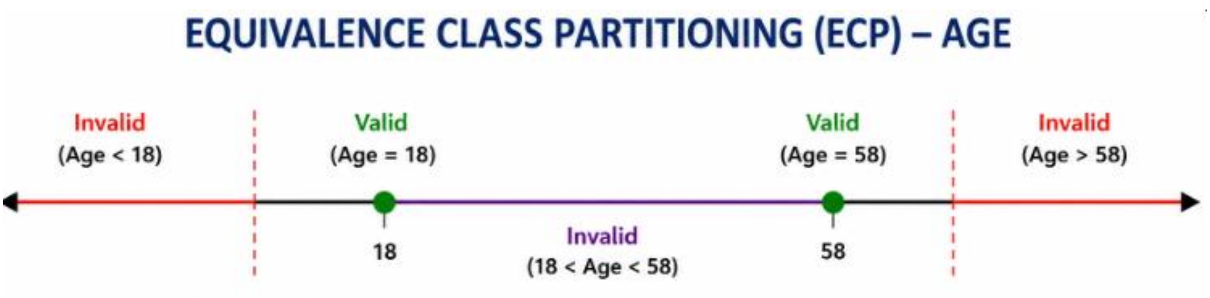
S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	**Partition 1**	Percentage ≤ 50 → **Invalid**	**40**
2	**Partition 2**	Percentage = 50 → **Valid**	**50**
3	**Partition 3**	Percentage 50–90 → **Valid**	**70**
4	**Partition 4**	Percentage = 90 → **Valid**	**90**
5	**Partition 5**	Percentage ≥ 90 → **Invalid**	**95**

MOBILE NUMBER Enter Mobile No. *Must be 10 digits

EQUIVALENCE PARTITIONING		
Invalid	**Valid**	**Invalid**
987654321	9876543210	98765432109

S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	Partition 1	Mobile Number → Less than 10 digits → Invalid	987654321
2	Partition 2	Mobile Number → Exactly 10 digits → Valid	9876543210
3	Partition 3	Mobile Number → More than 10 digits → Invalid	98765432109
4	Partition 4	Mobile Number → Contains alphabets → Invalid	98765ABCDE
5	Partition 5	Mobile Number → Contains special characters → Invalid	98765@3210

Find out Equivalence class partitioning and boundary value analysis for insurance application which accepts username age 18 and 58



S.No	Normal Test Data	Divide values into Equivalence Classes	Test Data using ECP
1	Partition 1	Age < 18 → Invalid	17
2	Partition 2	Age = 18 → Valid	18
3	Partition 3	18 < Age < 58 → Invalid	30
4	Partition 4	Age = 58 → Valid	58
5	Partition 5	Age > 58 → Invalid	59

S.No	Conditions	Status
1	**BVA min**	
	min = 18	Passed
	min-1 = 17	Failed
	min+1 = 19	Failed
2	**BVA max**	
	max = 58	Passed
	max-1 = 57	Failed
	max+1 = 59	Failed

Case Study for Decision Table

Search 5 tables for decision table. Create condition based on corresponding actions.

Requirement Number					
Condition	**1**	**2**	**3**	**4**	**5**
User is authorized	F	T	T	T	T
Chemical is available	—	F	T	T	T
Chemical is hazardous	—	—	F	T	T
Requester is trained	—	—	—	F	T
Action					
Accept request			X		X
Reject request	X	X		X	

Conditions:

1. **User is authorized**
2. **Chemical is available**
3. **Chemical is hazardous**
4. **Requester is trained**

Actions:

- **Accept request**
- **Reject request**

DECISION TABLE

ID	CONDITIONS/ACTIONS	TEST CASE 1	TEST CASE 2	TEST CASE 3	TEST CASE 4	TEST CASE 5
Condition 1	Account Already Approved	T	T	T	T	F
Condition 2	OTP (One Time Password) Matched	T	T	F	F	X
Condition 3	Sufficient Money in the Account	T	F	T	F	X
Action 1	Transfer Money	Execute				
Action 2	Show a Message as 'Insufficient Amount'		Execute			
Action 3	Block The Transaction Incase of Suspicious Transaction			Execute	Execute	X

Conditions

1. **Account Already Approved**
2. **OTP (One Time Password) Matched**
3. **Sufficient Money in the Account**

Actions

- Transfer Money
- Show a Message as **"Insufficient Amount"**
- Block the Transaction as a **Suspicious Transaction**

Printer troubleshooter

		Rules							
Conditions	Printer does not print	Y	Y	Y	Y	N	N	N	N
	A red light is flashing	Y	Y	N	N	Y	Y	N	N
	Printer is unrecognised	Y	N	Y	N	Y	N	Y	N
Actions	Check the power cable			X					
	Check the printer-computer cable	X		X					
	Ensure printer software is installed	X		X		X		X	
	Check/replace ink	X	X			X	X		
	Check for paper jam		X		X				

Conditions

1. **Printer does not print**
2. **A red light is flashing**
3. **Printer is unrecognized**

Actions

- **Check the power cable**
- **Check the printer-computer cable**
- **Ensure printer software is installed**
- **Check/replace ink**
- **Check for paper jam**

	Rule							
	1	2	3	4	5	6	7	8
Email is valid	Y	Y	Y	Y	N	N	N	N
Email has not been registered	Y	Y	N	N	Y	Y	N	N
Password length is 8 characters or more	Y	N	Y	N	Y	N	Y	N
Be able to register, navigate to homepage	X	-	-	-	-	-	-	-
Unable to register, an error message will be shown	-	X	X	X	X	X	X	X

Conditions

1. **Email is valid** — Y/N
2. **Email has not been registered** — Y/N
3. **Password length is 8 characters or more** — Y/N

Actions

- **Be able to register and navigate to homepage**
- **Unable to register; an error message will be shown**

Conditions	1	2	3	4	5	6	7	8
Traffic lights On?	T	T	T	T	F	F	F	F
Priority signs installed?	T	T	F	F	T	T	F	F
Traffic officer present?	T	F	T	F	T	F	T	F
Actions								
Regulated by traffic lights	-	T	-	T	-	-	-	-
Regulated by priority signs	-	-	-	-	-	T	-	-
Regulated by traffic officer	T	-	T	-	T	-	T	-
Give way to traffic on your right								T

Conditions

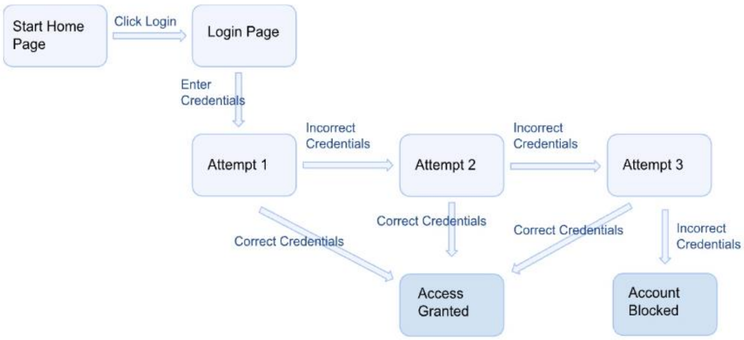
1. **Traffic lights On?**
2. **Priority signs installed?**
3. **Traffic officer present?**

Actions

- **Regulated by traffic lights**
- **Regulated by priority signs**
- **Regulated by traffic officer**
- **Give way to traffic on your right**

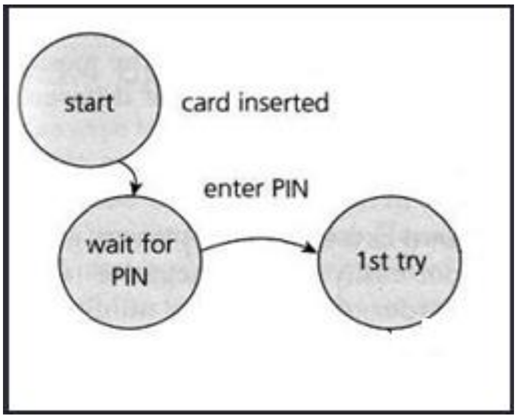
Case Study for State Transition Technique

Find out minimum 5 images from browser and identify state transition techniques.



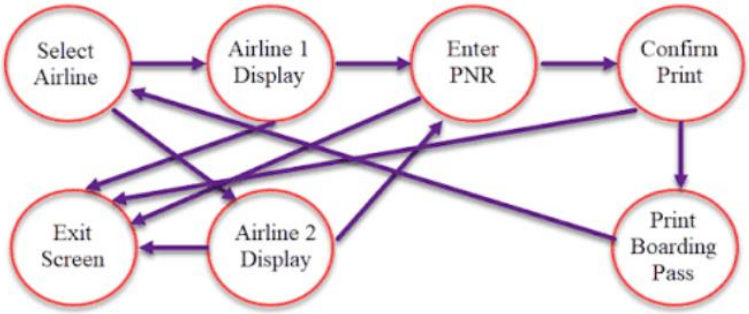
States: Home Page → Login Page → Attempt 1 → Attempt 2 → Attempt 3 → Access Granted / Account Blocked

1. **Valid Transition:** Correct credentials at any attempt → **Access Granted**
2. **Invalid Transition:** Incorrect credentials → **Next login attempt**
3. **Final Transition:** Three consecutive incorrect attempts → **Account Blocked**



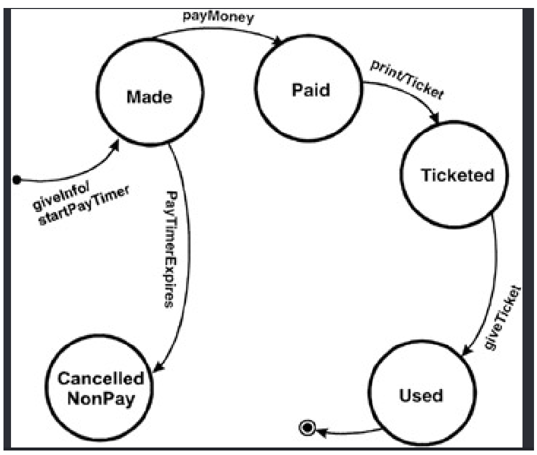
States: Start → Wait for PIN → 1st Try

1. **Valid Transition:** Card inserted → **Wait for PIN**
2. **Valid Transition:** Enter PIN → **1st Try**
3. **Input/Event:** Card inserted / Enter PIN → **Next State**



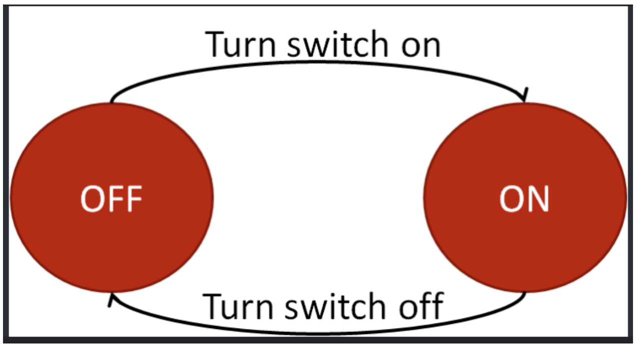
States: Select Airline → Airline 1 Display / Airline 2 Display → Enter PNR → Confirm Print → Print Boarding Pass / Exit Screen

1. **Valid Transition:** Select Airline → **Airline 1 Display / Airline 2 Display**
2. **Valid Transition:** Select Airline → Airline Display → **Enter PNR**
3. **Valid Transition:** Enter PNR → **Confirm Print**
4. **Valid Transition:** Confirm Print → **Print Boarding Pass**
5. **Exit Transition:** From the process → **Exit Screen**
6. **Alternative Transition:** Airline Display → **Select Airline / Exit Screen**



States: Made → Paid → Ticketed → Used / Cancelled NonPay

1. **Valid Transition:** GiveInfo / StartPayTimer → **Made**
2. **Valid Transition:** PayMoney → **Paid**
3. **Valid Transition:** PrintTicket → **Ticketed**
4. **Valid Transition:** GiveTicket → **Used**
5. **Invalid/Cancelled Transition:** PayTimer Expires → **Cancelled NonPay**
6. **Final State:** Used → **End**



States: OFF → ON

1. **Valid Transition:** Turn switch on → **ON**
2. **Valid Transition:** Turn switch off → **OFF**
3. **State 1:** Switch is **OFF**
4. **State 2:** Switch is **ON**